

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Dynamic Composite Load Model Template

WECC CMPLDW

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 Introduction

This document describes the WECC dynamic composite load model (CMPLDW) template that was delivered with previous *PowerFactory* versions (*PowerFactory* 2020, *PowerFactory* 2021, *PowerFactory* 2022 and *PowerFactory* 2023). It explains how to use and configure the model template.

Note: With *PowerFactory* 2024, the CMPLDW template is superseded by a built-in *PowerFactory* model of the WECC dynamic composite load model.

With *PowerFactory* 2024, the CMPLDW template is superseded by a built-in *PowerFactory* implementation of the WECC dynamic composite load model. For usual applications, it is recommended to use the built-in *PowerFactory* model, which is the *General Load* element (ElmLod) with *WECC Composite Load Type - CMPLDW* (TypLodwecc). Consequently, the CMPLDW template was removed from the global *DIgSILENT* templates library. Nevertheless, dynamic models of components of the CMPLDW template still exist in the *DIgSILENT* dynamic models library, because the superseded template can still be in use in *PowerFactory* projects of users.

The CMPLDW template represents the WECC dynamic composite load model based on the document published by WECC [1] in a modular way. Individual components of the CMPLDW model are represented by individual network element (e.g. the 3-phase motors, the single-phase air conditioner, the electronic load, static load, feeder impedance, etc.). This gives the possibility to access the load bus, which might be of interest for special studies. However, it also requires a more complex configuration procedure of the model, as explained in Chapter 3, 4 and 5, and has an impact on the simulation performance (speed).

Note: The superseded WECC dynamic composite load model (CMPLDW) template is still available in the *DIgSILENT* Knowledge Base for download, if required for special studies. Search for “Do you have a modular implementation of the WECC dynamic composite load model (CMPLDW)?” within the *DIgSILENT* Knowledge Base:
https://www.digsilent.de/en/faq-powerfactory.html

The WECC dynamic composite load model (CMPLDW) template can be used for:

- Load Flow Studies
- Stability Studies (RMS Simulation with positive sequence network representation)

For dynamic simulation a variable step size is possible. The minimum step size or the fix step size should be lower than 20ms.

2 How to Insert the Template into a Network Model

Inserting the templates is simple, just click on the *General Templates* button ( )and select the desired template in the following window. Thereafter you could place the model in the single line diagram. For this you have to click one time in the single line diagram, this shows the model connected to the mouse pointer a second click creates a new model of the selected template in the single line diagram. Each dynamic model consists of some elements in the single line diagram (mostly a terminal with a generator) plus a composite model, which is located in the recording grid.

After you placed all models in the single line diagram you have to close the selection window with the templates, this terminates the template inserting.

You could delete a placed model directly by pressing the undo button ( ) or you could delete the model also manually. If you want to delete a model manually you have to mind that the model consists of visible elements in the single line diagram and also of a composite model which is only visible in the Data Manager.

3 Model Description

The composite load model is represented by static generators in the single line diagram. The static generators are controlled by the currents in d- and q- axis [2]. They represent different load types, namely:

- Three 3-phase double-cage induction motors (Motor A, B, C)
- One 1-phase air conditioner performance-based model (Motor D)
- Electronic load model
- Static load model

Figure 3.1 shows the network model of the composite load model. The dynamic models of the loads are collected in the composite model *WECC Composite Load Model*. This composite model could be found either through the link *WECC Composite Load Model* on the basic data page of the different loads or with the Data Manger in the used grid.

The composite load model also includes two static generators(Bfdr1, Bfdr2) which act as compensators in order to minimise the difference between the given reactive power setpoint by the user and the actual generated reactive power. This is done using the station controller which controls these two static generators during the load flow. These compensators are necessary as the reactive power generated by the 3-phase induction motors can not be known in advance i.e. function of the active power and the slip. Hence, these compensators, which act as constant impedances, correct the load flow and do not have any influence on the dynamic performance of the load model.

Moreover, the active power setpoint is distributed using the corresponding shares of the each load, hence no adjustment is required in this case as the active power consumption is known before hand.

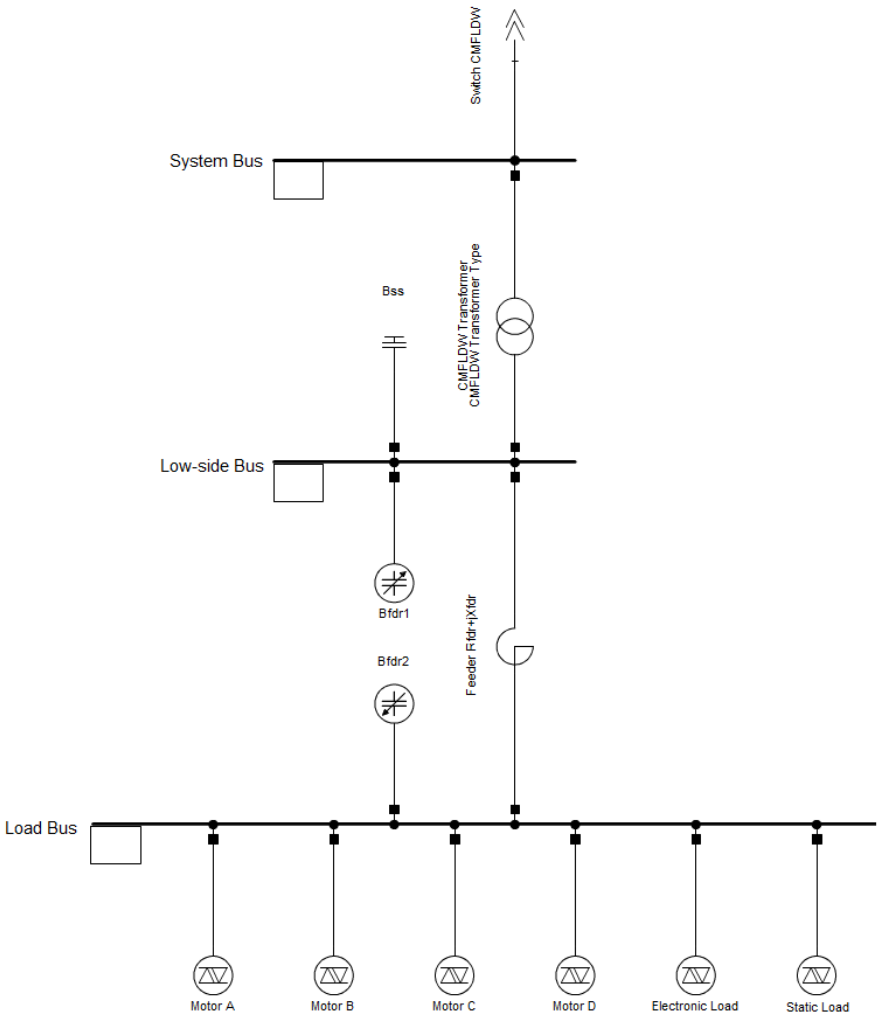


Figure 3.1: Network representation *WECC Composite Load*

The composite load model has been created in a folder which is named as *WECC CMPLDW*. The composite load model consists of the following elements:

- Dynamic model of the transformer tap changer (Transformer Tap Changer)
- Dynamic model of the composite load model (WECC Composite Load Model)
- Composite load model configuration script (WECC CMPLDW Configuration)
- QDSL models for the steady state load flow initialisation of 3-phase induction motors(Motor Motor A/B/C Steady-State Calculation)
- QDSL model for the calculation of the reactive power setpoint of the1-phase air conditioner performance-based model (motor D)
- QDSL model for the active power control of static load

The configuration script is used to set the active, reactive power, the share of the different load types, Base MVA, nominal voltages of buses and transformer, equivalent feeder resistance and

reactance as well the transformer tap changer and impedances. Figure 3.2 shows the basic options page of the configuration script.

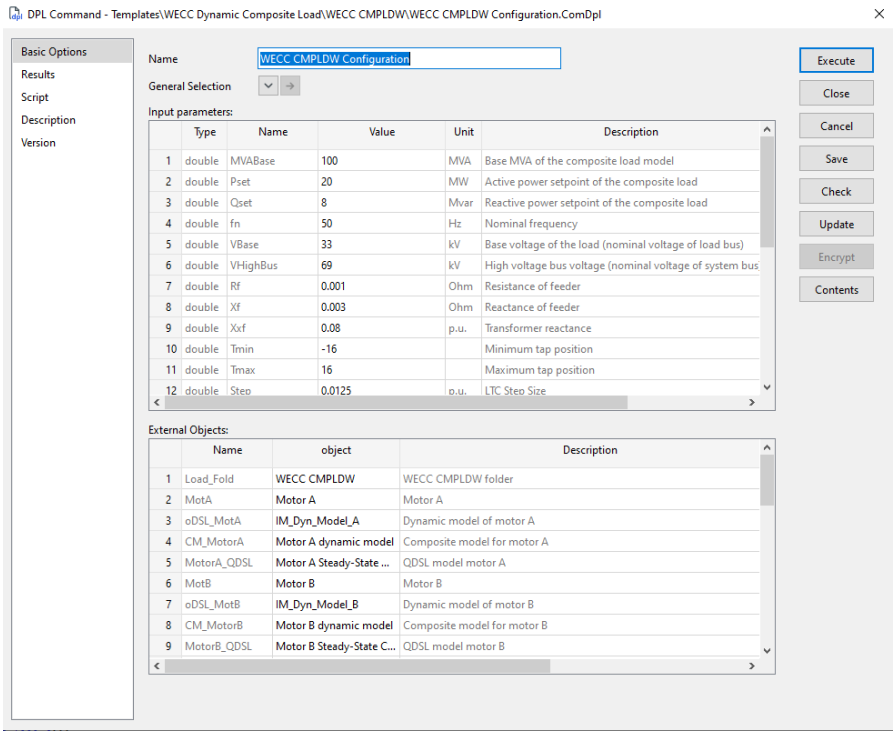


Figure 3.2: Composite load model configuration script

The initial speeds of the 3-phase induction motors are calculated by the QDSL models. The calculated speeds from the load flow are used by the configuration script for initialisation embedded inside the block definition of dynamic model of the 3-phase induction motors to initialise the state variables of the motors. For the calculation of the Vstallbreak voltage for motor D, a similar configuration script for the initialisation is used which is linked with the dynamic model of motor D.

The equations used for the calculation of active and reactive power of the 1-phase A/C performance-based model (Motor D) in running states are given below

if $V > V_{brk}$

$$P = [Po + Kp1 \cdot (V - Vbrk)^{Np1}] \cdot [1 + CmpKpf \cdot \Delta f] \quad (1)$$

$$Q = [Qo + Kq1 \cdot (V - Vbrk)^{Nq1}] \cdot [1 + CmpKqf \cdot \Delta f] \quad (2)$$

if $V < V_{brk}$ and $V > Vstallbrk$

$$P = [Po + Kp2 \cdot (Vbrk - V)^{Np2}] \cdot [1 + CmpKpf \cdot \Delta f] \quad (3)$$

$$Q = [Qo + Kq2 \cdot (Vbrk - V)^{Nq2}] \cdot [1 + CmpKqf \cdot \Delta f] \quad (4)$$

if $V < Vstallbrk$

$$P = 1/Rstall \cdot V^2 \quad (5)$$

$$Q = 1/Xstall \cdot V^2 \quad (6)$$

where:

- $Kp1$ and $Kp2$ are active power coefficients for running states 1 and 2 respectively
- $Np1$ and $Np2$ are active power exponents for running states 1 and 2 respectively
- $Kq1$ and $Kq2$ are reactive power coefficients for running states 1 and 2 respectively
- $Nq11$ and $Nq2$ are reactive power exponents for running states 1 and 2 respectively
- $CmpKpf$ is active power frequency sensitivity
- $CmpKqf$ is reactive power frequency sensitivity

Further information on the modelling of the 1-phase A/C performance-based model (Motor D) can be found in [1].

4 How to Configure the Model

This chapter describes how to configure and setup the model.

The load flow and network data can be entered in the following DPL script:

1. DPL configuration script ”WECC CMPLDW Configuration.ComDpl”

The dynamic data of each load type can be entered in the following models:

1. Dynamic model of motor A ”IM_Dyn_Model_A.ElmDsl”
2. Dynamic model of motor B ”IM_Dyn_Model_B.ElmDsl”
3. Dynamic model of motor C ”IM_Dyn_Model_C.ElmDsl”
4. Dynamic model of motor D ”Motor D dynamic model.ElmDsl”
5. Dynamic model of static load ”Static load dynamic model.ElmDsl”
6. Dynamic model of electronic load ”Electronic load dynamic model.ElmDsl”

Changing the active and reactive power setpoints

In order to change the active and reactive power setpoints, the Pset and Qset values shown in Figure 3.2 should be changed. Based on these setpoints, the model will be initialised. The Qset and Pset are the values at the HV side of the composite load model transformer.

Changing the nominal power

In order to change the nominal power, the MVABase value should be changed as shown in Figure 3.2.

Changing the nominal AC voltage

The nominal AC voltages of the composite load model can also be changed using the configuration script shown in Figure 3.2.

Changing share of the loads in the composite load model

The share of the 3-phase induction motors, 1-phase A/C performance-based model (Motor D) and electronic load can be changed in the input window of the configuration script as shown in

Figure 3.2. If the sum of the fractions of the motors and the electronic load is larger than 1, then the fractions are normalised to the sum of 1. If the sum of the fractions of the motors and the electronic load is less than 1, then the reminder is represented by the static load model.

Changing the dynamic data of the composite load model

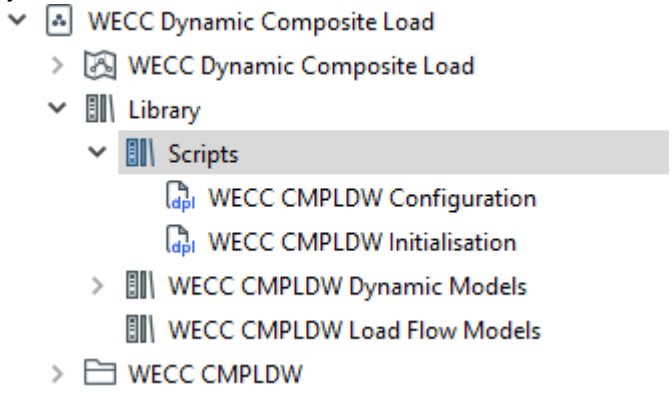
The dynamic data of the different load models can be changed in the respective common models of the different loads.

The contactor and the thermal relay of Motor D have been implemented as arrays, hence their parameters are not visible in the common model of the motorD. In order to change and modify the settings of the contactor and thermal relay, the user should click on the ”Advanced 1”page of the Motor D dynamic model. There, the parameters of the contactor and thermal protection can be varied.

5 How to Use the model

In order to insert the template into the grid and initialise it, the below steps must be followed:

1. Insert as many templates in the grid as required (see Chapter 2).
2. The composite load models will be inserted as folders “WECC CMPLDW” in the grid. Go to each folder, open the “WECC CMPLDW Configuration” script within the folder (in the network model) and change the active / reactive power setpoints, nominal AC voltages, base MVA, share of different load types as required by the user. Moreover, the dynamic data of the load model can also be changed in the respective common models. Refer to Chapter 4 for more details.
3. Execute the configuration script “WECC CMPLDW Configuration” directly from within the “WECC CMPLDW” folder in the network model (for each inserted CMPLDW) or use the initialisation script “WECC CMPLDW Initialisation” which can be found in the template library as shown below:



4. Run the RMS simulation.

The advantage of the WECC initialisation script is that it searches for the WECC_CMPLDW folders in the active grid and executes the WECC CMPLDW Configuration script for each WECC load model one by one. After the execution of all the configuration scripts, the initialisation script performs the load flow thus making sure that all the inserted models are initialised correctly.

This template contains also one transformer type which is included in the composite dynamic load model. The transformer type can be found in the folder named “Transformer Type CMPLDW” which is a subfolder of the “WECC CMPLDW” folder. Each load has its own transformer type so that each load can be configured individually.

Note: The names of the folders in which composite load models are created should not be changed. This is only relevant for the “WECC CMPLDW Initialisation” script. Users who do not intend to run the initialisation script, may change the names of the composite load model folders. In that case, the user has to execute the individual configuration scripts for each inserted load models.

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6 References

[1] WECC Composite Load Model Specification, January 2015.

[2] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.

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